Validated Synthetic Patient Generation for Small Longitudinal Cohorts: Coagulation Dynamics Across Pregnancy

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Abstract

Small longitudinal clinical cohorts—common in maternal health, rare diseases, and early-phase trials—are often too small for robust computational modeling or machine learning. Generating synthetic patients that faithfully capture the joint, longitudinal, and mechanistic structure of real data would enable analyses that are otherwise infeasible. Here, we present a synthetic patient generation pipeline based on multiplicity-weighted Stochastic Attention (SA), a sampling framework rooted in modern Hopfield network theory that extends the standard Hopfield energy with per-pattern multiplicity weights, enabling continuous interpolation between unconditioned generation and targeted subpopulation amplification at inference time without retraining. We applied this framework to a longitudinal coagulation dataset collected from K=23 pregnant patients across three clinical visits (pre-pregnancy baseline, first trimester, and third trimester). By concatenating multi-visit patient profiles into unified patterns and sampling via Langevin dynamics on the multiplicity-weighted energy landscape, SA generated coherent longitudinal trajectories that preserved the geometry of the clinical data manifold. We validated synthetic patients through a four-level framework of increasing stringency: (1) marginal plausibility of individual features, (2) preservation of cross-visit covariance structure, (3) amplification of rare clinical subgroups (e.g., polycystic ovary syndrome, n=3 $\rightarrow$ 100) with preserved condition-specific signatures, and (4) mechanistic consistency with an independent 58-species ordinary differential equation model of the coagulation cascade. At all four levels, SA-generated patients were indistinguishable from real patients (86–94% cloud overlap under mechanistic validation), while a multivariate normal baseline failed at levels 2–4 due to its rank-deficient covariance estimate (n<p). These results demonstrate that geometry-preserving generative methods can produce clinically validated synthetic cohorts from very small longitudinal datasets.

Keywords: synthetic data generation, stochastic attention, modern Hopfield networks, multiplicity weighting, coagulation, pregnancy, longitudinal data, mechanistic validation

1 Introduction

Pregnancy induces profound changes in the coagulation and fibrinolytic systems, shifting the hemostatic balance toward a hypercoagulable state [Grouzi et al., 2022]. Pregnant and postpartum women face a 4–5 fold increased risk of venous thromboembolism compared with non-pregnant women, and between 2017 and 2019, thrombotic embolisms caused 8.7% of pregnancy-related deaths in the United States, while hemorrhage caused 13.7% [Trost et al., 2022, Harris, 2023]. Understanding patient-specific coagulation dynamics during pregnancy is therefore critical for risk stratification and clinical decision-making. However, certain pregnancy complications—polycystic ovary syndrome (PCOS), preeclampsia (PE), and gestational thrombophilia—are sufficiently rare that assembling adequately powered longitudinal cohorts is prohibitively expensive and slow. Typical cohort sizes are on the order of tens of patients with complete longitudinal follow-up, yielding datasets where the number of features p exceeds the number of patients n. This $n < p$ regime fundamentally limits conventional statistical and machine learning approaches: covariance matrices are rank-deficient, cross-validation is unreliable, and overfitting is nearly inevitable.

Synthetic data generation offers a path forward by augmenting small real datasets with statistically faithful synthetic records, enabling downstream computational analyses that would otherwise be infeasible. The most straightforward approach is sampling from a fitted multivariate normal (MVN) distribution, which captures first- and second-order statistics by construction. However, MVN has three critical limitations in the small-cohort longitudinal setting. When $n < p$, the sample covariance matrix is singular and must be regularized, introducing bias that distorts the joint distribution [Ledoit and Wolf, 2004]. In our dataset (K=23 patients, p=216 concatenated features), the MVN covariance has rank at most 22, leaving the remaining 194 dimensions unconstrained. Beyond this structural deficiency, MVN assumes Gaussian marginals and linear correlations, which are poor approximations for clinical measurements with skewed or heavy-tailed distributions. Finally, MVN provides no mechanism for conditional generation—it cannot amplify a specific clinical subpopulation (e.g., the n=3 PCOS patients) while preserving subgroup-specific signatures. More expressive generative models such as GANs and VAEs [Goodfellow et al., 2014, Kingma and Welling, 2014] require substantially larger training sets and are prone to mode collapse when n is as small as 23.

Stochastic Attention (SA) provides an alternative framework rooted in modern Hopfield network theory [Ramsauer et al., 2021]. Modern Hopfield networks define an energy landscape over continuous state vectors where stored patient profiles serve as memory patterns, and by running Langevin dynamics on this energy landscape, SA generates novel samples that interpolate between stored patterns in a manner governed by the attention mechanism. We recently introduced a multiplicity-weighted extension of the Hopfield energy that adds per-pattern weights r_k to the log-sum-exp term, enabling continuous interpolation between unconditioned generation and targeted subpopulation amplification at inference time without retraining the energy landscape [Varner, 2024]. Crucially, SA operates on the data manifold directly rather than fitting a parametric distribution, making it well-suited to the $n < p$ regime where parametric approaches must regularize or truncate the high-dimensional structure of the data. In prior work, we generated synthetic patient populations using multivariate normal distributions fitted independently to each clinical visit [Bravo et al., 2023]. While this approach captured per-visit statistics, it assumed Gaussian marginals and generated each visit independently rather than as part of a coherent longitudinal trajectory.

In this work, we adapted multiplicity-weighted SA for longitudinal clinical data and introduced a four-level validation framework to rigorously assess synthetic patient quality. We formulated a pipeline for applying SA to continuous multivariate longitudinal measurements, including concatenated-visit patterns that treat each patient's full longitudinal profile as an atomic unit, PCA dimensionality reduction, and a direction-magnitude decomposition that preserves the anisotropic variance structure lost by unit-norm projection. The multiplicity-weighted energy provided a unified mechanism for both unconditioned generation (all weights equal) and targeted amplification of rare clinical subgroups (designated patterns upweighted), without requiring separate models or retraining. We validated the resulting synthetic patients through four progressively stringent levels: marginal plausibility of individual features, preservation of cross-visit covariance structure, amplification of rare clinical subgroups with preserved condition-specific signatures, and mechanistic consistency with an independent 58-species ODE model of the coagulation cascade. We demonstrated that SA-generated synthetic patients passed all four validation levels when applied to a 72-feature, 3-visit pregnancy coagulation dataset, while MVN generation failed at levels 2 through 4, and that an independent mechanistic model confirmed 86–94% cloud overlap between real and synthetic patients under physiological conditions.

2 Methods

2.1 Clinical Dataset

We analyzed a longitudinal dataset of coagulation, fibrinolysis, thrombin generation assay (TGA), and rotational thromboelastometry (ROTEM) measurements collected from pregnant patients at three clinical visits: Visit 1 (pre-pregnancy baseline), Visit 2 (first trimester), and Visit 3 (third trimester). The dataset contained 153 total records across approximately 50 subjects. After removing columns with $> 30\%$ missing values and retaining only subjects with complete data across all three visits, we obtained $K = 23$ complete longitudinal profiles across $n = 72$ assay measurements per visit.

The 72 assay measurements spanned five categories: (i) hormones (estradiol, progesterone), (ii) coagulation factors and inhibitors (Factors II, V, VII, VIII, IX, X, XI, XII; AT, PC, TFPI, vWF, fibrinogen, etc.), (iii) fibrinolytic markers (plasminogen, PAI-1, PAI-2, α_2-AP, TAFI, tPA, D-dimer equivalents), (iv) thrombin generation parameters under four initiator conditions (TF, TF+TM, No TF, No TF+anti-FXIa), and (v) ROTEM/viscoelastic parameters (CT, MCF, alpha angle, LOT, ML, LT, AUC under TF Only and TF+tPA conditions). The dataset included three clinical subgroups: Healthy (n=14), polycystic ovary syndrome (PCOS, n=3), and preeclampsia (PE, n=5), with the remaining patient having other complications.

2.2 Multiplicity-Weighted Stochastic Attention

We generated synthetic patients using a multiplicity-weighted variant of Stochastic Attention (SA) rooted in modern Hopfield network theory [Ramsauer et al., 2021]. We stored K memory patterns as columns of a matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d\times K}$ and defined a weighted Hopfield energy landscape:

$$E_r(\boldsymbol{\xi}) = \frac{1}{2}\,\|\boldsymbol{\xi}\|^2 - \frac{1}{\beta}\log\sum_{k=1}^{K} r_k \exp\left(\beta\,\mathbf{m}_k^\top \boldsymbol{\xi}\right) \tag{1}$$

where $\{\mathbf{m}_k\}_{k=1}^K$ are stored memory patterns, β is an inverse temperature parameter controlling the sharpness of retrieval, and $\{r_k\}_{k=1}^K$ are per-pattern multiplicity weights. When all $r_k = 1$, this reduces to the standard modern Hopfield energy; when $r_k = \rho > 1$ for a designated subset of patterns, the energy landscape is continuously deformed to favor that subset. We sampled from this landscape using an Unadjusted Langevin Algorithm (ULA):

$$\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\hat{\mathbf{X}}\,\mathrm{softmax}\Big(\beta\,\hat{\mathbf{X}}^\top\boldsymbol{\xi}_t + \log\mathbf{r}\Big) + \sqrt{\frac{2\alpha}{\beta}}\,\boldsymbol{\varepsilon}_t \tag{2}$$

where α is a step size, $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\log \mathbf{r}$ is the element-wise log of the multiplicity vector. The deterministic component drove $\boldsymbol{\xi}_t$ toward a multiplicity-weighted combination of stored patterns, while the stochastic component enabled exploration and novelty. The $\log r_k$ bias in the softmax logits provided a minimal, theoretically grounded mechanism for continuously interpolating between unconditioned generation ($\rho = 1$, all patterns equally weighted) and hard subset curation ($\rho \to \infty$, only designated patterns contribute). This is an inference-time operation that requires no retraining of the energy landscape.

The inverse temperature β controlled the trade-off between retrieval (large β, converging to nearest memory) and generation (small β, uniform mixing). We identified the critical β^* at the phase transition by computing the normalized attention entropy and finding the inflection point (maximum negative second derivative of $\bar{H}(\beta)$ in log-β space) over a logarithmic sweep. For weighted sampling, the entropy calculation incorporated the multiplicity bias, yielding a ρ-dependent $\beta^*(\rho)$. We operated near β^* to produce samples that were novel yet statistically consistent with the stored patterns.

The effective number of patterns contributing to the dynamics was quantified by the participation ratio $K_{\text{eff}} = (\sum_k r_k)^2 / \sum_k r_k^2$, which interpolated smoothly between the total pattern count (unconditioned) and the designated subset size (strongly conditioned). For a target effective fraction f_{target} of probability mass on the designated subset, the required multiplicity was $\rho = f_{\text{target}} \cdot K_{\text{bg}} / (K_{\text{des}} \cdot (1 - f_{\text{target}}))$, where K_{des} and K_{bg} are the designated and background pattern counts, respectively.

2.3 Pipeline for Continuous Longitudinal Data

Longitudinal concatenation. For each of the $K = 23$ patients, we concatenated the $n = 72$ measurements across all three visits into a single vector of dimension $d_{\text{concat}} = 3n = 216$, ensuring that generated synthetic patients had internally consistent longitudinal trajectories.

Standardization and PCA. We standardized each feature to zero mean and unit variance, then applied PCA retaining 95% of the variance, reducing dimensionality from 216 to $d_{\text{PCA}} = 18$. This yielded a memory-to-dimension ratio of $K/d_{\text{PCA}} \approx 1.28$, placing SA in a favorable operating regime where the number of stored patterns exceeded the dimensionality of the representation.

Direction-magnitude decomposition. Standard SA operates on unit-norm patterns, which destroys the anisotropic variance structure of continuous data. We addressed this by recording PCA-space norms $\{r_k\}$ before normalization, running SA on unit-norm patterns to obtain a direction $\hat{\boldsymbol{\xi}}$, drawing a magnitude from the empirical distribution of norms, and reconstructing $\boldsymbol{\xi}_{\text{rescaled}} = r \cdot \hat{\boldsymbol{\xi}}$. The rescaled vector was mapped back to feature space via inverse PCA and destandardization, then split into three 72-dimensional visit records.

Conditional generation. To generate condition-specific cohorts (e.g., PCOS-only), we set the multiplicity $r_k = \rho$ for the designated patterns and $r_k = 1$ for all others, and sampled using the weighted Langevin update (Eq. 2). The multiplicity ρ was chosen to achieve a target effective fraction of probability mass on the designated subset. Magnitudes for the direction-magnitude decomposition were drawn from the condition-specific subset of norms, ensuring that the scale of variation matched the target subpopulation.

2.4 Multivariate Normal Baseline

As a baseline, we generated synthetic patients by fitting a multivariate normal distribution to the same K=23 concatenated patient profiles (p=216). Because $K < p$, the sample covariance matrix had rank at most 22 and was regularized using Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004].

2.5 Validation Framework

We evaluated synthetic patient quality through four progressively stringent levels. **Level 1 (Marginal Plausibility)** tested whether individual feature distributions matched, using per-feature means, standard deviations, and known physiological relationships. **Level 2 (Joint Structure)** tested whether the cross-visit covariance was preserved, by comparing full 216×216 correlation matrices, eigenvalue spectra, and PCA projections between real, SA, and MVN populations. **Level 3 (Conditional Structure)** tested whether rare subgroups could be amplified while preserving condition-specific signatures, by generating cohorts of 100 patients each from the Healthy (n=14), PCOS (n=3), and PE (n=5) subgroups using multiplicity-weighted sampling. **Level 4 (Mechanistic Consistency)** tested whether synthetic patients produced biologically plausible outputs under an independent mechanistic model. We used the Hockin–Mann BZ2012 coagulation model [Hockin et al., 2002, Luan et al., 2007], a system of 58 ODEs with 64 rate constants, in which 5 rate constants were calibrated on Visit 1 real patients at the population level and the remaining 59 were held at literature values. We ran the model on all real and synthetic patients under two conditions (TF-only and TF+TM) and compared predicted-versus-measured thrombin generation features, using cloud overlap—the fraction of synthetic predicted/measured ratios falling within the 5th–95th percentile range of real ratios—as the primary metric.

3 Results

We generated N=100 synthetic longitudinal patient profiles using SA and compared them against the K=23 real patients across four progressively stringent validation levels. As a baseline, we also generated N=100 patients from a regularized multivariate normal (MVN) distribution fitted to the same data.

3.1 Marginal Plausibility

We first assessed whether SA-generated patients reproduced per-feature summary statistics across all three visits. The median mean relative error (MRE) across all 72 features was approximately 2%, with the majority of features below 5% (Table 1), indicating that SA faithfully captured the central tendencies of the real population. Representative coagulation factors—including Factor II (MRE $<$ 1.5%), Factor VIII (MRE $<$ 3%), antithrombin (MRE $<$ 2%), and fibrinogen (MRE $<$ 1.2%)—showed particularly strong agreement across all visits. MVN also performed well at this level (median MRE $\approx$ 2–3%), which was expected since moment-matching is guaranteed by construction.

Table 1: Per-visit mean relative error (MRE) between real and SA-generated synthetic patients for representative coagulation features, computed as $|\bar{x}_{\text{synth}} - \bar{x}_{\text{real}}|/|\bar{x}_{\text{real}}|$.

Feature	Visit 1	Visit 2	Visit 3
Factor II (%)	0.009	0.012	0.015
Factor VIII (%)	0.017	0.030	0.025
AT (%)	0.013	0.018	0.014
Fibrinogen (mg/dL)	<0.001	0.008	0.012
TF Peak (nM)	0.022	0.019	0.027
TF ETP (nM·min)	0.015	0.021	0.018

We next examined whether known physiological relationships were preserved in the synthetic cohort. We plotted pairwise scatter plots of coagulation factor levels against thrombin generation parameters across all visits and observed that SA-generated patients occupied the same joint regions as real patients (Fig. 1). The expected inverse relationship between antithrombin and TF-initiated peak thrombin was preserved, as were the positive associations between Factor VIII and peak, prothrombin and ETP, and the inverse relationship between antithrombin and ETP. These results indicated that SA captured not only univariate distributions but also the biologically meaningful pairwise dependencies that underlie coagulation physiology.

We also evaluated whether the characteristic pregnancy-driven longitudinal trajectories were reproduced. We computed per-visit means and standard deviations for six key coagulation features and found that synthetic patients tracked the real trajectories closely, with overlapping confidence bands at all three visits (Fig. 2). Fibrinogen, Factor VIII, and von Willebrand factor showed the expected monotonic increases from baseline through the third trimester in both cohorts, while Factor II increased modestly and antithrombin declined then partially recovered, consistent with the known hypercoagulable shift of pregnancy. These patterns confirmed that the concatenated-visit SA pipeline preserved the directionality and magnitude of longitudinal change.

3.2 Cross-Visit Covariance Structure

We compared the full 216×216 cross-visit correlation matrices for real, SA-generated, and MVN-generated populations. The real data exhibited a characteristic block structure with strong within-visit correlations along the diagonal and structured off-diagonal blocks reflecting cross-visit dependencies—for example, a patient's Visit 1 Factor X level predicting their Visit 3 Factor X level. We found that SA preserved this block structure, including the off-diagonal cross-visit correlations (Fig. 3). The SA−Real residual matrix showed small, unstructured deviations distributed throughout, whereas the MVN−Real residual was notably smoother, reflecting the regularization-induced shrinkage of off-diagonal correlations toward zero. This difference was most apparent in the off-diagonal blocks, where MVN systematically underestimated cross-visit dependencies that SA retained.

We examined the eigenvalue spectrum of the sample covariance matrix to understand the structural origin of this difference. The real data had rank 22 (eigenvalues dropping to numerical zero at component 23), reflecting the $K{=}23$ patient constraint (Fig. 4). SA's spectrum dropped at component 18, the PCA truncation point, faithfully reflecting the low-rank structure without introducing variance in null dimensions. In contrast, MVN's Ledoit–Wolf regularization inflated eigenvalues beyond component 22, producing spurious variance in dimensions where the data contained none. This spurious variance manifested as the increased dispersion visible in PCA projections: SA-generated patients occupied the same region of PCA space as real patients across all three visits, while MVN-generated patients showed substantially greater scatter, particularly at Visits 2 and 3 (Fig. 5).

3.3 Conditional Generation of Rare Subgroups

We tested whether SA could amplify rare clinical subpopulations while preserving their condition-specific signatures. Our dataset contained three clinical subgroups—Healthy ($n{=}14$), PCOS ($n{=}3$), and PE ($n{=}5$)—of which the PCOS and PE groups were too small for any independent statistical analysis. We used SA's multiplicity-weighted sampling to generate condition-specific cohorts of 100 patients each by upweighting attention on the relevant stored patterns.

We projected all real and generated patients into PCA space and observed that the three synthetic cohorts separated consistently with the real between-group differences, with synthetic

Healthy, PCOS, and PE patients clustering around their respective real counterparts (Fig. 6). This separation was maintained across all three individual visits (Supplementary Fig. S3), indicating that condition-specific structure was preserved not only in aggregate but at each pregnancy timepoint. Condition-specific feature means were well-preserved: the Healthy cohort showed a median MRE of approximately 4%, while the smaller PCOS cohort had a median MRE of approximately 8%, reflecting the greater uncertainty inherent in amplifying from only three stored patterns. These results suggested that SA's attention-weighted interpolation on the data manifold can meaningfully amplify rare subgroups in a way that preserves their distinguishing clinical features—a capability with direct relevance to hypothesis generation and power analysis in understudied patient populations. MVN has no mechanism for this: fitting a separate MVN to three PCOS patients across 216 features is mathematically impossible (rank 2 covariance), and post-hoc subsetting of unconditional MVN samples does not preserve subgroup-specific structure.

3.4 Mechanistic Consistency

As a final, orthogonal validation, we tested whether SA-generated patients produced biologically plausible outputs when their coagulation factor levels were fed through an independent mechanistic model of the coagulation cascade. We used the Hockin–Mann BZ2012 model [Hockin et al., 2002, Luan et al., 2007], a system of 58 ordinary differential equations with 64 rate constants that simulates thrombin generation from patient-specific coagulation factor inputs. We calibrated 5 of the 64 rate constants on Visit 1 real patients at the population level (not per-patient), holding the remaining 59 at published literature values. We then ran the calibrated model on all 23 real patients (3 visits $\times$ 2 conditions = 138 simulations, 0 failures) and all 100 synthetic patients (3 visits $\times$ 2 conditions = 600 simulations, 0 failures), producing predicted thrombin generation features (lagtime, peak, time-to-peak, maximum rate, and ETP) that we compared against the corresponding measured values.

Under TF-only conditions, we found that synthetic patients produced the same predicted-versus-measured cloud as real patients across all five TGA features (Fig. 7). The median predicted-to-measured ratio was 0.92$\times$ for synthetic patients versus 0.91$\times$ for real patients, with median absolute relative errors of 19.4% and 21.1%, respectively. We quantified this overlap by computing the fraction of synthetic predicted/measured ratios that fell within the 5th–95th percentile range of the real ratios and found cloud overlap ranging from 86% for lagtime to 94% for maximum rate (Fig. 8), which indicated that the SA-generated factor-to-TGA mapping was indistinguishable from that of real patients at the population level.

Under TF+TM conditions, the model systematically overcorrected the protein C anticoagulant pathway, producing peak and maximum rate predictions approximately 0.5$\times$ measured values for both real and synthetic patients (Supplementary Fig. S4). This shared systematic bias was itself informative: if synthetic patients had violated the underlying coagulation biology, they would have deviated from real patients in how the mechanistic model processed their factor levels. Instead, cloud overlap under TF+TM remained high at 88–91%.

We also verified that the calibration generalized across pregnancy timepoints. The model was calibrated only on Visit 1 real patients, yet the predicted-to-measured ratios at Visits 2 and 3 under TF-only conditions were 0.96$\times$ and 1.07$\times$, respectively (Supplementary Fig. S6), confirming that the calibrated rate constants captured a stable population-level property of the coagulation system rather than overfitting to the training visit. Spearman rank correlations between predicted and measured values were moderate for ETP ($\rho \approx 0.6$–0.8 for real patients, $\rho \approx 0.6$–0.65 for synthetic; Supplementary Fig. S7) and weak for other features, which was expected given that the Hockin–Mann model was not designed to resolve inter-patient variability from a 5-parameter population-level calibration. The key finding was not patient-level prediction but population-level plausibility: an independent mechanistic model, encoding decades of coagulation biochemistry and knowing nothing about the SA generation process, confirmed that synthetic patients fell within the same biologically plausible envelope as real patients.

4 Discussion

The four-level validation framework provided complementary evidence that SA-generated synthetic patients faithfully represented the real clinical population. Level 1 confirmed that marginal distributions were preserved—a necessary but insufficient condition that MVN also satisfied by construction. Level 2 demonstrated that SA preserved the joint cross-visit covariance structure that MVN's rank-deficient estimate could not faithfully represent, a difference made visible in the eigenvalue spectrum where MVN introduced spurious variance in null dimensions while SA respected the low-rank geometry of the data. Level 3 showed that SA could amplify clinically meaningful rare subgroups—generating 100 synthetic PCOS patients from only 3 real ones—while preserving subgroup-specific signatures that MVN fundamentally could not capture. Level 4 provided orthogonal mechanistic evidence: an independent ODE model, encoding decades of coagulation biochemistry and calibrated exclusively on real patients, confirmed that the factor-to-thrombin-generation mapping in synthetic patients was biologically plausible, with 86–94% cloud overlap under TF-only conditions. Taken together, these results established that SA-generated synthetic patients were not merely statistically similar to real patients but were consistent with the underlying biology at multiple levels of resolution.

The ability to generate validated synthetic cohorts from very small longitudinal datasets has immediate practical implications for maternal health research. Rare pregnancy complications such as PCOS, preeclampsia, and gestational thrombophilia are difficult to study because assembling large, well-characterized longitudinal cohorts requires years of recruitment across multiple clinical sites. Our results suggested that SA can enable hypothesis generation, power analyses, and computational modeling studies for these understudied populations by generating synthetic cohorts that preserve both the statistical and mechanistic properties of the real data. The conditional generation capability demonstrated at Level 3 is particularly relevant: rather than requiring researchers to collect additional rare patients, SA can amplify existing small subgroups into cohorts large enough for meaningful analysis.

The key insight underlying SA's advantage over parametric alternatives is geometric. SA generates samples on the data manifold by interpolating between stored patient profiles via the Hopfield energy landscape, rather than fitting a parametric distribution that must regularize or truncate high-dimensional structure. In the $n < p$ regime that characterizes most small clinical studies, any parametric distribution—whether Gaussian or otherwise—faces a fundamental identifiability problem: there are more parameters to estimate than observations to estimate them from. SA sidesteps this by operating in a PCA-reduced space where the memory-to-dimension ratio is favorable ($K/d_{\mathrm{PCA}} \approx 1.28$) and by preserving the full directional structure of the data through the attention mechanism. The eigenvalue spectrum comparison (Fig. 4) made this distinction concrete: SA's spectrum faithfully reflected the rank-18 PCA structure, while MVN's regularized spectrum introduced artificial variance in 194 additional dimensions.

The mechanistic validation at Level 4 introduced a paradigm that we believe is broadly applicable beyond coagulation. Rather than relying solely on statistical comparisons between real and synthetic distributions, we tested whether synthetic patients produced biologically consistent outputs when processed by a domain-specific computational model. This approach is available whenever a validated mechanistic model exists for the domain of interest—pharmacokinetic models in drug development, physiological models in critical care, or metabolic models in oncology. The requirement is not that the mechanistic model be a perfect predictor (ours had weak rank correlations at the individual patient level), but that synthetic patients fall within the same model-predicted envelope as real patients. The shared systematic bias we observed under TF+TM conditions reinforced this interpretation: both real and synthetic patients were processed identically by the mechanistic model, producing the same pattern of overcorrection in the protein C pathway.

Several limitations warrant discussion. First, our study used a single longitudinal dataset of K=23 patients, and further evaluation on independent datasets and across different clinical domains is needed to establish generalizability. Second, the PCA dimensionality reduction assumed a linear structure in the feature space; nonlinear manifold learning methods may better capture complex clinical relationships, though at the cost of reduced interpretability. Third, the mechanistic validation under TF+TM conditions revealed systematic model bias, limiting the interpretability of Level 4 results under that condition to the observation that real and synthetic patients shared the same bias pattern. Fourth, we did not validate synthetic patients against clinical outcomes—our validation was restricted to statistical and mechanistic plausibility, and prospective validation against real clinical decision-making remains an important future direction.

5 Conclusion

We have demonstrated that Stochastic Attention can generate synthetic longitudinal patient data from very small clinical cohorts (K=23) that pass four levels of validation: marginal plausibility, cross-visit covariance preservation, conditional subgroup amplification, and mechanistic consistency with an independent ODE model of the coagulation cascade. The SA framework preserves the geometry of the clinical data manifold in regimes where parametric approaches such as multivariate normal sampling fail due to rank deficiency. Combined with a rigorous multi-level validation framework, this approach provides a practical path for generating clinically validated synthetic cohorts to support computational research in rare conditions and small longitudinal studies.

Data Availability

The code for synthetic patient generation, all validation scripts, and the generated synthetic datasets are available at `https://github.com/varnerlab/SA-generation-legacy-dataset-paper`. The real patient data were collected under IRB approval at the University of Vermont Medical Center; de-identified data are available upon reasonable request to the corresponding author.

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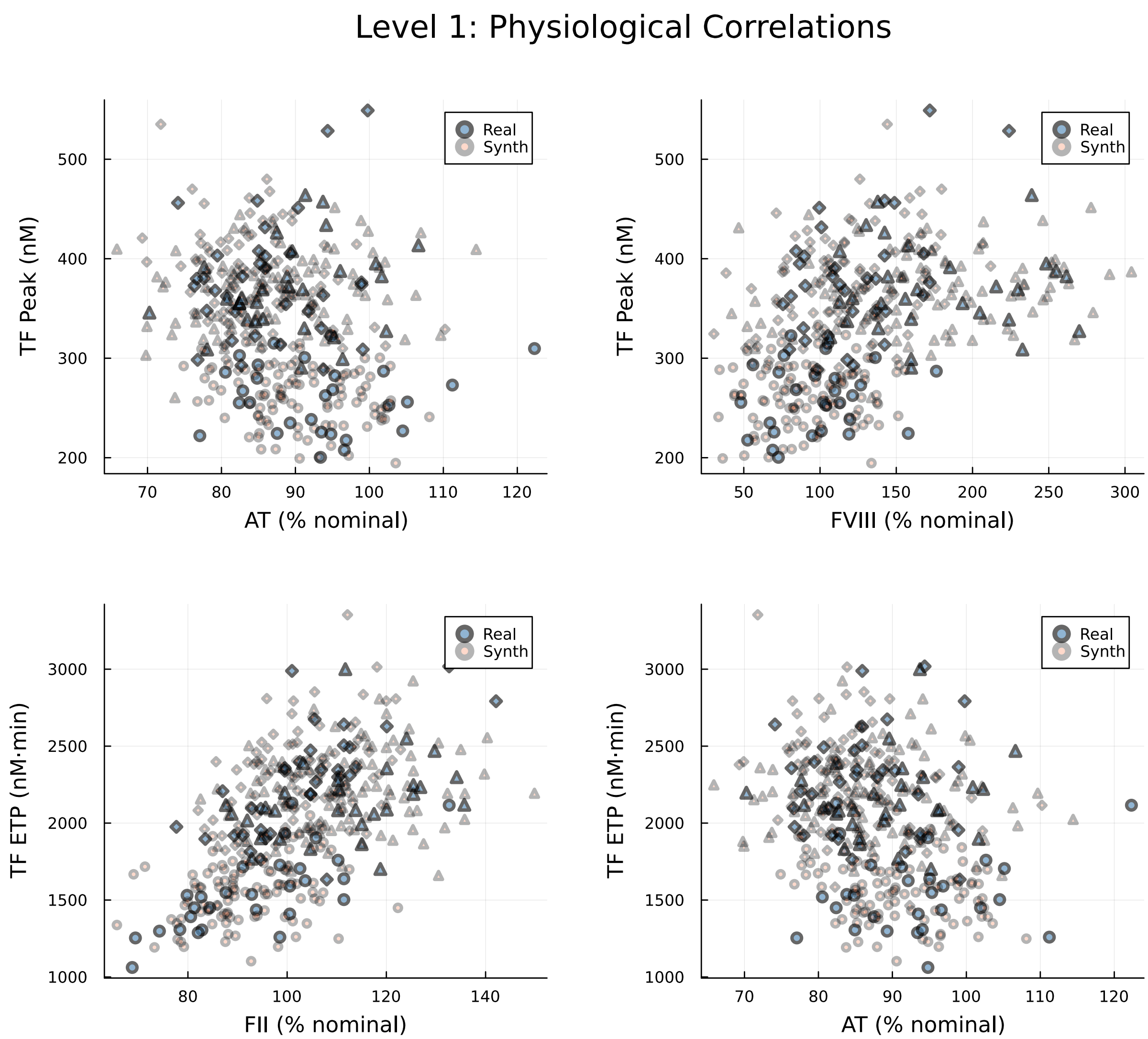

Figure 1: **Physiological correlations are preserved in synthetic patients.** Scatter plots of coagulation factor levels versus thrombin generation parameters across all visits (marker shapes indicate visit). Real patients (blue) and SA-generated synthetic patients (tan) occupy the same joint regions, confirming that biologically meaningful pairwise dependencies—including the AT–peak inverse relationship (top left), the FVIII–peak positive relationship (top right), and the FII–ETP and AT–ETP associations (bottom)—are preserved.

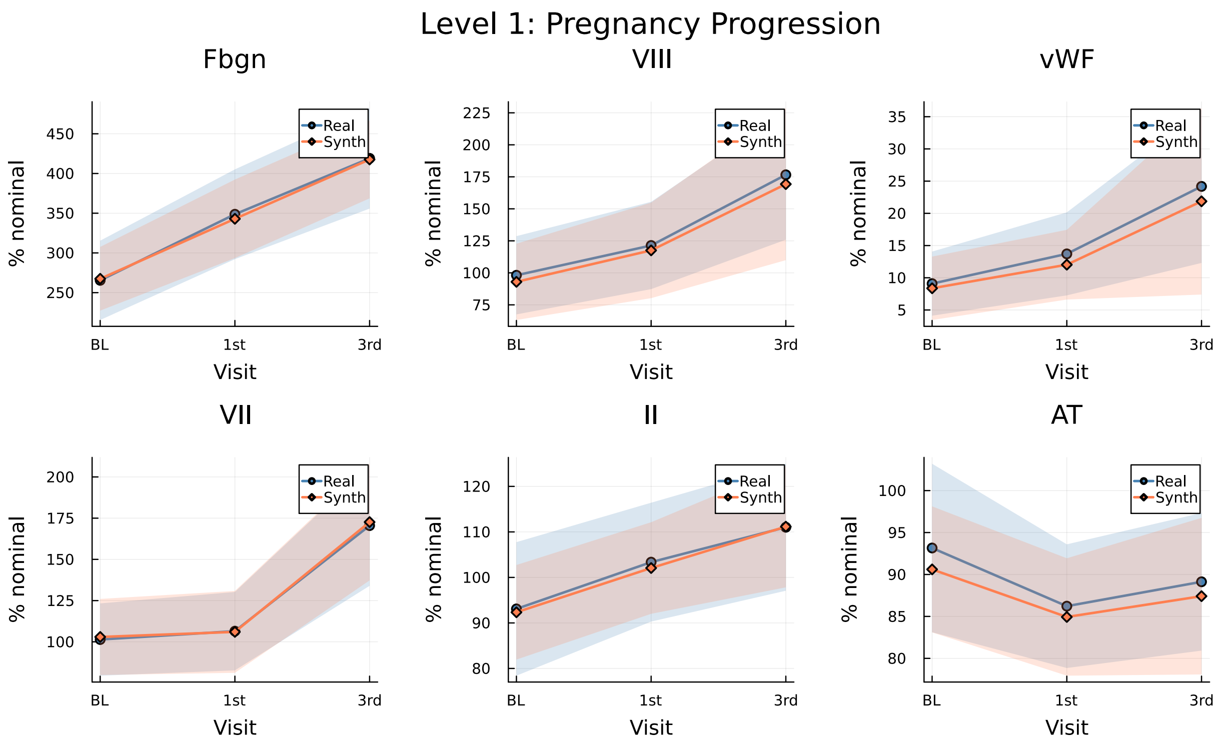

Figure 2: **Pregnancy-driven longitudinal trajectories are faithfully reproduced.** Mean ± standard deviation bands for six key coagulation features across visits (BL = baseline, 1st = first trimester, 3rd = third trimester) for real (blue) and SA-generated synthetic (orange) patients. The characteristic pregnancy-driven increases in fibrinogen, Factor VIII, and vWF are captured, as are the stable-to-declining patterns in Factor II and AT.

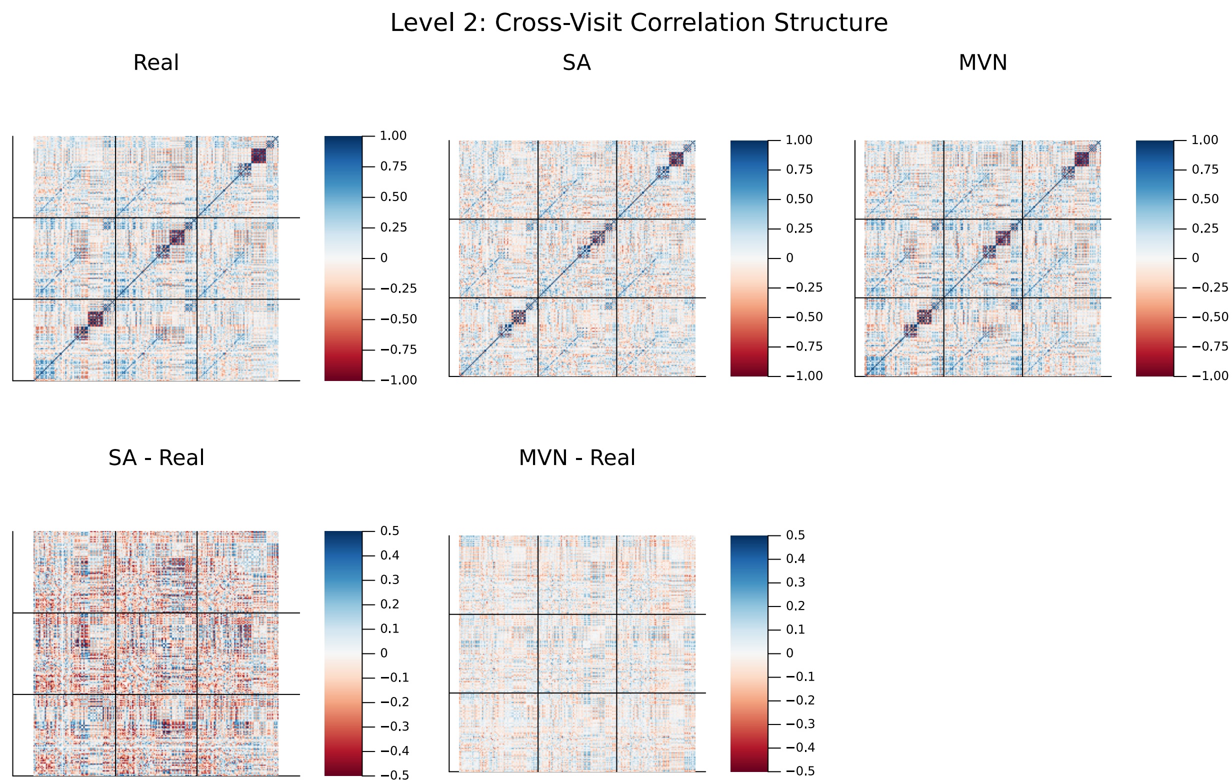

Figure 3: **Cross-visit correlation structure.** Top row: full 216×216 Pearson correlation matrices for Real (left), SA (center), and MVN (right) populations, with black lines delineating the three 72×72 visit blocks. Bottom row: residual matrices (SA−Real and MVN−Real). SA preserves the off-diagonal block structure encoding cross-visit dependencies; MVN's regularized estimate shrinks these correlations toward zero.

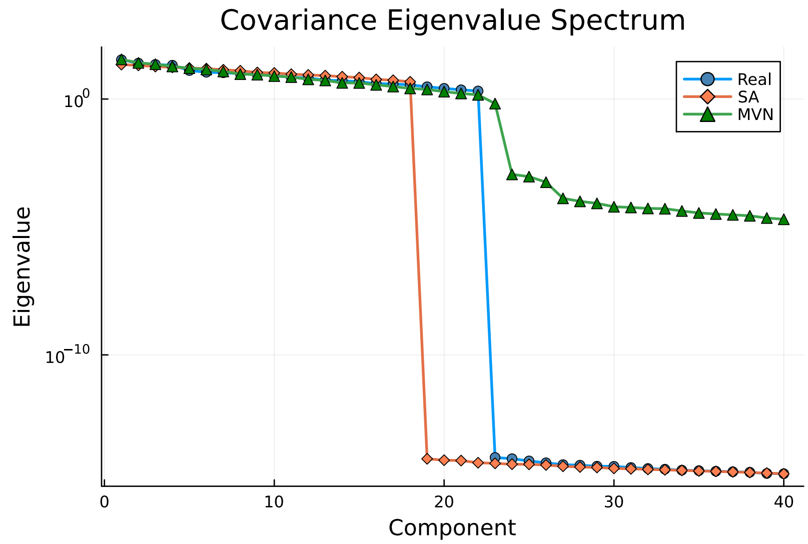

Figure 4: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216 × 216 sample covariance for Real (K=23, rank 22), SA (N=100), and MVN (N=100) populations. SA's spectrum drops at component 18 (the PCA truncation), matching the low-rank data structure. MVN's Ledoit–Wolf regularization inflates eigenvalues beyond component 22, introducing spurious variance in null dimensions.

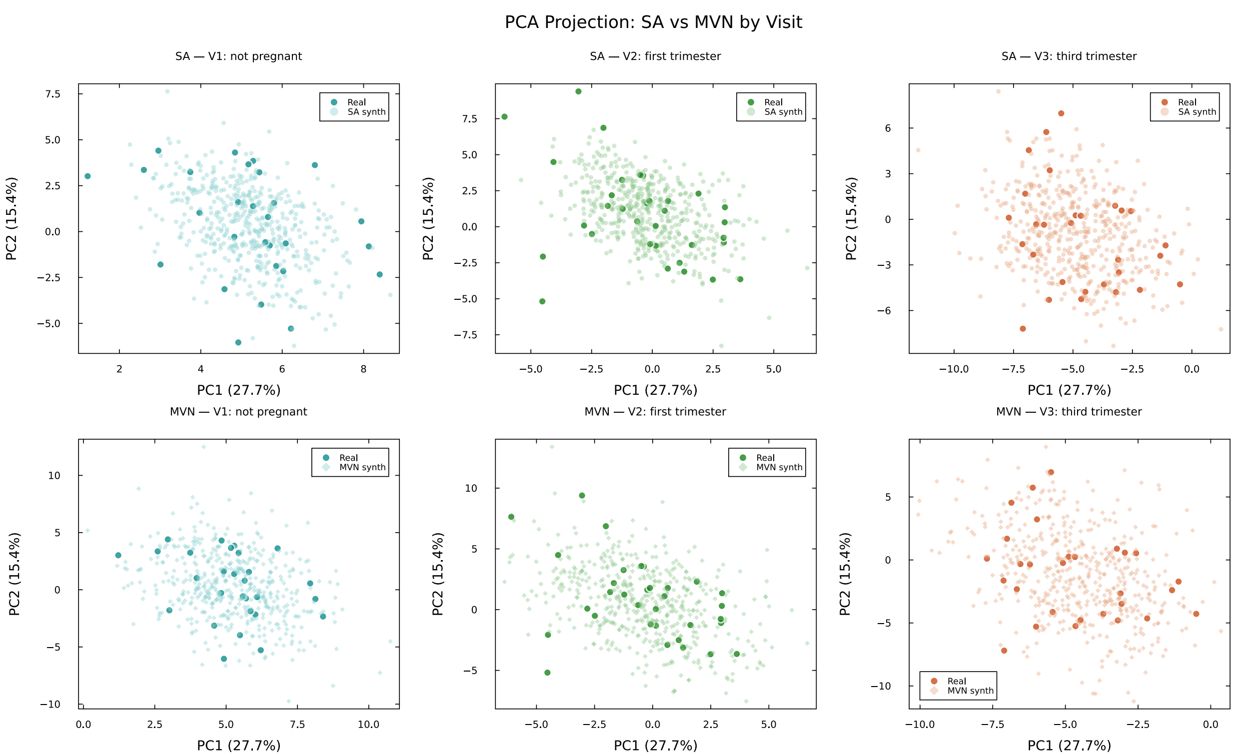

Figure 5: **PCA projections by visit.** Top row: SA-generated patients (light) overlaid on real patients (dark) for each visit. Bottom row: MVN-generated patients overlaid on real patients. SA patients concentrate around the real data cloud at all visits, while MVN patients show greater dispersion, particularly at Visits 2 and 3.

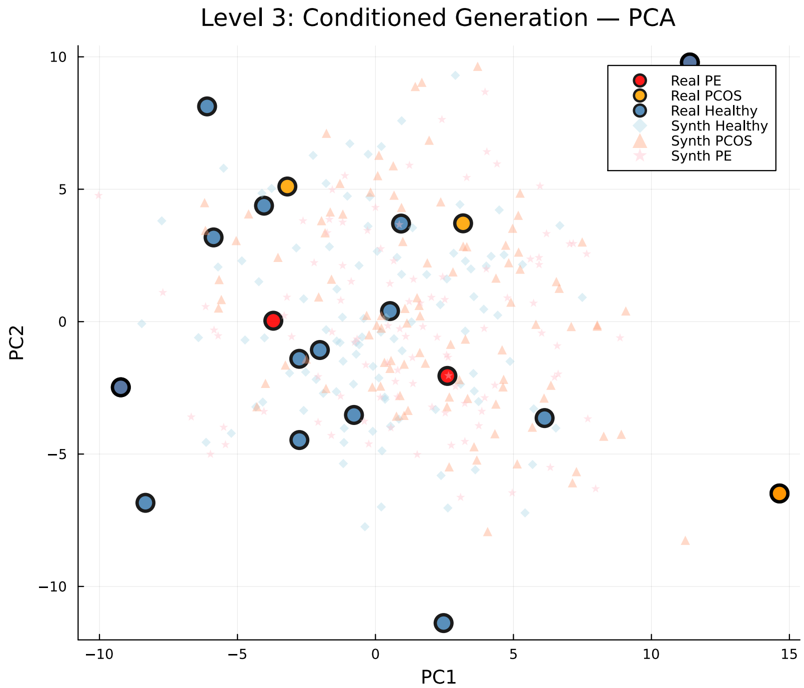

Figure 6: **Conditional generation of rare clinical subgroups.** PCA projection of condition-specific synthetic cohorts generated by multiplicity-weighted SA. Large markers: real patients (blue = Healthy, n=14; orange = PCOS, n=3; red = PE, n=5). Small markers: synthetic patients (N=100 per condition). Each synthetic cohort clusters around its corresponding real patients, demonstrating that SA preserves condition-specific signatures even when amplifying from as few as three stored patterns.

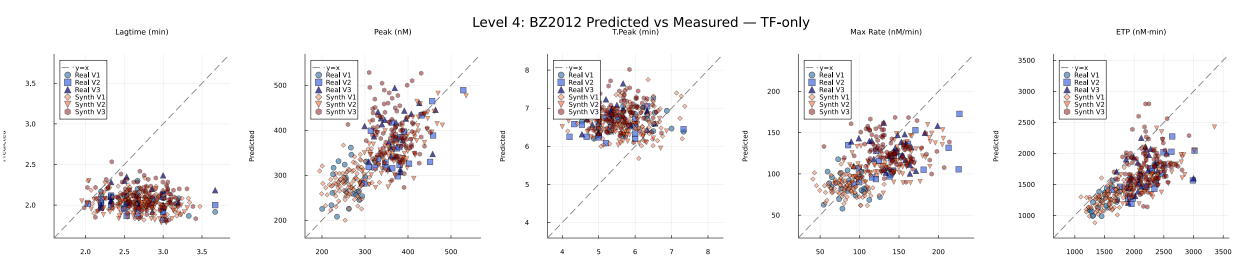

Figure 7: **Mechanistic model predicted versus measured TGA features (TF-only).** Each panel shows one TGA feature; the dashed line indicates perfect prediction (y=x). Blue markers: real patients (circles = V1/train, squares = V2, triangles = V3). Red/brown markers: SA-generated synthetic patients (diamonds = V1, inverted triangles = V2, hexagons = V3). Synthetic patients occupy the same predicted-versus-measured cloud as real patients across all features and visits.

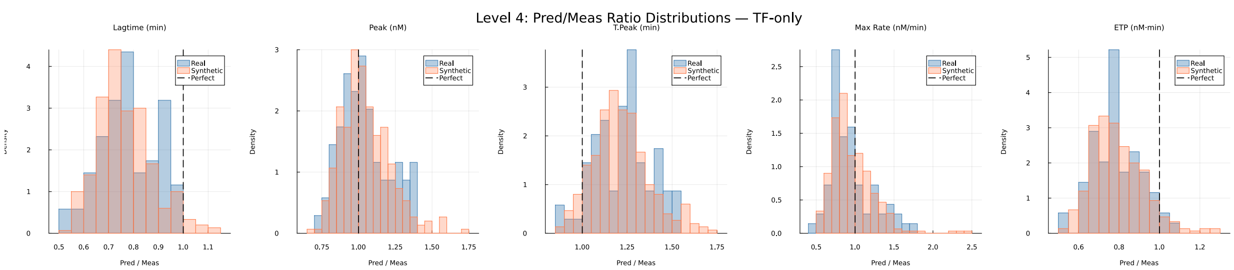

Figure 8: **Predicted/measured ratio distributions (TF-only).** Histograms of the predicted-to-measured ratio for each TGA feature for real (blue) and SA-generated synthetic (orange) patients. The dashed line indicates perfect prediction (ratio = 1). The distributions overlap substantially, with cloud overlap ranging from 86% (lagtime) to 94% (maximum rate).